

BHUBANESHWAR, IndiaWMO: **429710**Lat: **20.244N**Lon: **85.818E**Elev: **42**StdP: **100.82**Time Zone: **5.50 (IND)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	13.8	14.9	7.8	6.6	25.1	9.4	7.4	23.9	6.0	23.7	5.1	23.4	1.0	0	0.357

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	9.9	39.3	27.3	38.0	27.2	36.8	26.9	29.7	34.8	29.2	34.0	28.9	33.4	4.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.6	25.2	31.9	28.2	24.5	31.4	28.0	24.2	31.1	98.8	34.7	96.7	34.5	95.0	33.3	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.2	7.2	11.7	43.1	1.0	1.6	11.0	44.3	10.4	45.2	9.8	46.1	9.1	47.3		
			10.5	30.9	1.1	0.9	9.6	31.5	9.0	32.1	8.4	32.6	7.5	33.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	22.3	25.8	29.3	31.3	32.2	30.7	28.8	28.7	28.6	27.6	24.9	22.2
	DBStd	3.63	1.93	2.30	1.75	1.85	2.01	2.43	1.45	1.21	1.28	1.40	1.74	1.72
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6458	381	441	599	639	687	620	583	579	558	546	447	378
	CDD18.3	3417	123	208	341	389	428	370	325	321	308	287	197	120
	CDH23.3	38425	850	1829	3795	5037	5956	4878	3795	3649	3459	2874	1522	779
	CDH26.7	17519	221	809	1883	2781	3560	2566	1452	1316	1256	1006	494	176
(14) Wind	WSAvg	2.7	1.7	2.2	3.2	4.9	4.7	3.5	3.0	2.7	2.2	1.7	1.6	1.4
(15) Precipitation	PrecAvg	1514	14	20	27	32	66	197	373	366	239	160	32	3
	PrecMax	1931	88	83	98	151	261	493	799	747	466	533	180	48
	PrecMin	905	0	0	0	0	1	26	180	110	0	28	0	0
	PrecStd	280	23	22	30	36	61	113	125	135	100	122	44	9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.0	37.0	39.0	40.9	41.7	41.9	35.2	34.1	34.2	33.9	32.6	30.9
		MCWB	19.7	20.9	23.8	26.4	27.2	28.6	28.6	28.4	28.1	27.0	23.7	21.2
	2%	DB	30.2	34.6	37.0	38.8	39.2	38.4	33.9	33.1	33.3	33.0	31.5	29.7
		MCWB	20.2	21.4	23.9	26.6	28.4	28.4	28.4	28.2	27.9	26.3	23.1	20.3
	5%	DB	29.0	33.0	35.7	37.4	37.9	36.4	32.9	32.2	32.6	32.1	30.6	28.6
		MCWB	19.9	21.7	24.3	26.7	28.3	28.4	28.1	28.0	27.7	26.0	22.9	19.7
	10%	DB	27.8	31.4	34.4	36.0	36.6	34.8	31.8	31.4	31.7	31.2	29.6	27.5
		MCWB	19.4	21.4	24.4	26.7	28.0	28.3	27.9	27.8	27.6	25.8	22.6	19.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.7	25.7	28.2	29.7	30.5	30.5	29.5	29.2	29.2	28.5	26.5	23.8
		MCDB	28.0	30.5	33.7	35.9	37.0	36.0	33.2	32.3	32.8	31.9	30.0	28.2
	2%	WB	22.5	24.9	27.1	28.7	29.7	29.6	29.0	28.7	28.6	27.8	25.6	22.5
		MCDB	26.6	29.1	32.7	35.4	35.5	34.5	32.3	31.7	32.0	30.9	28.8	26.3
	5%	WB	21.7	24.2	26.5	28.1	29.3	29.2	28.5	28.3	28.2	27.3	24.9	21.4
		MCDB	26.0	28.1	31.5	34.0	34.8	33.7	31.5	31.1	31.3	30.3	28.0	25.4
	10%	WB	20.9	23.5	26.1	27.6	28.9	28.8	28.1	28.0	27.8	26.8	24.2	20.5
		MCDB	25.1	27.4	30.3	32.8	34.1	33.0	30.9	30.6	30.5	29.6	27.3	25.0
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	12.4	11.5	10.8	9.9	7.5	5.4	5.3	5.8	7.3	9.8	11.7
		MCDBR	13.2	13.9	13.0	12.2	11.2	10.4	7.2	6.6	7.3	8.5	10.6	12.3
		MCWBR	4.3	3.9	3.6	3.3	3.3	3.2	2.9	2.8	2.9	3.2	3.6	4.3
	5% WB	MCDBR	10.7	11.0	10.8	10.6	10.0	8.7	6.3	6.0	6.4	6.9	8.1	10.0
		MCWBR	4.4	3.6	3.4	3.5	3.5	3.6	3.0	2.9	3.0	3.2	3.4	4.2
(40) Clear-Sky Solar Irradiance	taub		0.546	0.537	0.605	0.606	0.650	0.683	0.596	0.565	0.567	0.569	0.575	0.578
	taud		1.843	1.896	1.785	1.830	1.768	1.707	1.908	2.004	1.991	1.938	1.840	1.790
	Ebn at Noon		721	763	730	735	697	668	730	756	749	727	691	674
	Edh at Noon		193	193	224	216	228	240	196	179	179	183	192	198
(44) All-Sky Solar Radiation	RadAvg		4.32	5.19	5.88	6.46	6.30	4.75	3.95	4.16	4.41	4.62	4.34	3.98
	RadStd		0.17	0.23	0.25	0.24	0.29	0.45	0.39	0.32	0.34	0.43	0.26	0.29

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.17	N/A	N/A	+0.98	+0.40	+0.30	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	+0.40	+0.33	N/A	N/A	N/A	N/A

Nomenclature: See separate page